Data Preprocessing Report

Feature engineering and preprocessing of the Loan Prediction dataset

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Dataset	Loan Prediction Problem (Kaggle)	Date	October 2026

1. Summary

The Loan Prediction training file (614 applications, 13 columns) was inspected, cleaned, enriched with new features, encoded, scaled and reduced to a final set of **8 features plus the target**. Every value the pipeline learns from the data (fill values, outlier limits, scaling averages, feature-selection statistics) comes from the **training rows only**; 20% of the data was set aside before any fitting, so the hold-out set stays a fair test. The result is a dataset with no missing values, fully numeric, free of redundant features, and reproducible from a single pipeline that also works on unseen applications.

Item	Result
Rows and columns (raw / final)	614 x 13 / 614 x 10 (8 features, target, split flag)
Train / hold-out split	491 / 123 rows, stratified (approval rate 68.6% / 69.1%)
Missing values	7 columns affected; all handled; 0 remain
Duplicates	0 found
New features created	11 (affordability ratio, total income, household size, log versions)
Features kept	8 of 24 candidates
Cross-validated ROC-AUC: raw vs prepared data	0.643 vs 0.732 (selected set)

2. Data inspection

The first ten rows, structure, data types, summary statistics, duplicates and missing values were examined in the notebook. Key findings:

- **Structure:** one identifier (Loan_ID), eleven predictors and the target. Five columns are numeric, eight are text.
- **Duplicates:** none, whether or not Loan_ID is ignored.
- **Skewness:** applicant income (6.54), co-applicant income (7.49) and loan amount (2.68) are strongly right-skewed.
- **Odd storage:** Dependents is text containing "3+", and Credit_History is a number that really represents a category.
- **Target imbalance:** 68.7% approved against 31.3% rejected.

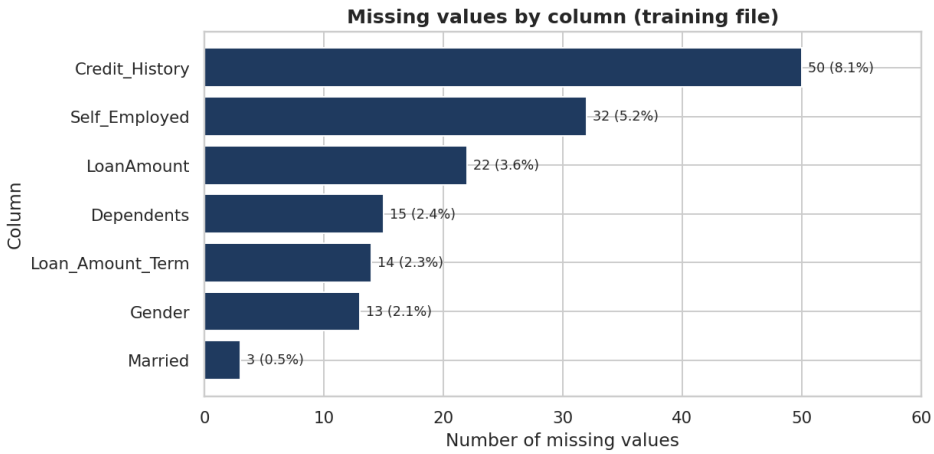


Figure 1. Missing values by column. Credit_History is the largest gap (50 values, 8.1%); 134 of 614 rows (21.8%) have at least one gap.

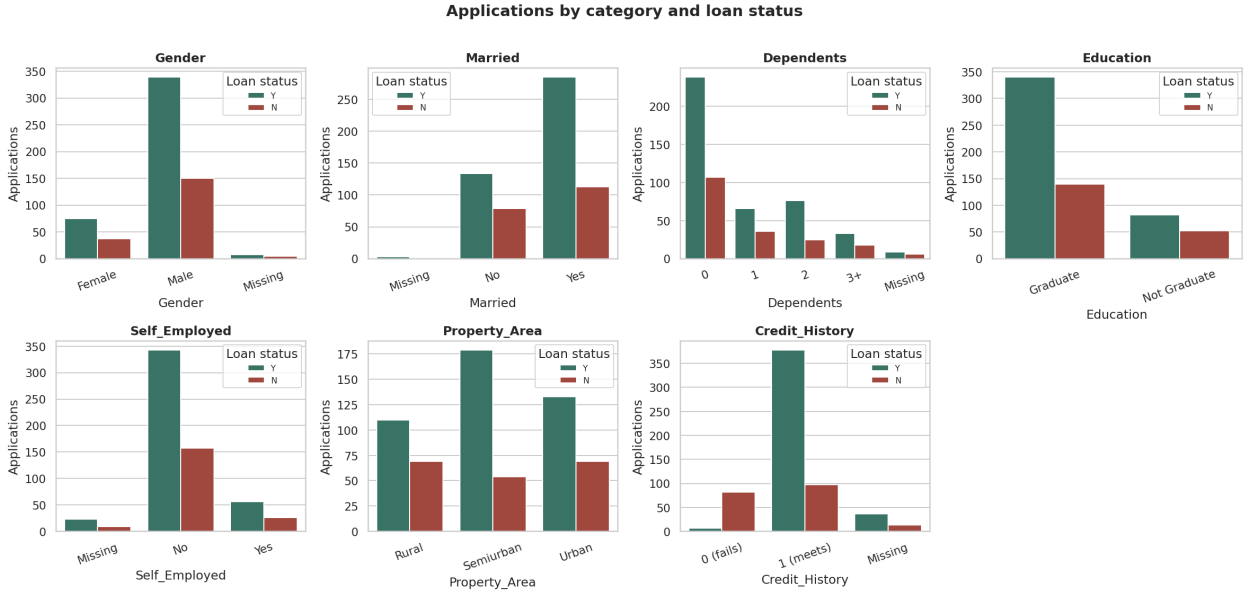


Figure 2. Applications by category and loan status. Credit history shows the clearest difference between approved and rejected.

3. Preprocessing decisions

3.1 Data cleaning

Dependents "3+" became the number 3; credit history 1/0 became the labels Meets/Fails; text was trimmed; columns were renamed to clear snake_case names with units (for example loan_amount_k is in thousands). The duplicate check was repeated after these corrections and again found nothing.

3.2 Missing values

Each column received the treatment that suits its meaning. The fill values were learned from the training rows and then applied to every row.

Column	Missing	Treatment	Fill value	Justification
gender	13	Mode	Male	Few gaps; one category dominates
married	3	Mode	Yes	Very few gaps
self_employed	32	Mode	No	86% of applicants answered No
dependents	15	Mode	0	A count where 0 is by far the most common
loan_amount_k	22	Median	128	Skewed amounts: the median is not pulled up by huge loans
loan_term_months	14	Mode	360	83% of loans have a 360-month term
credit_history	50	New category	Unknown	Strongest predictor: guessing would invent information, and "no record" is a real situation

3.3 Feature engineering

New features were built from lending logic: how much income is available, how large the repayment is against it, and how many people it supports.

New feature	How it is built	Why it helps
total_income	applicant + co-applicant income	Repayment ability depends on household income
has_coapplicant	1 if co-applicant income is above 0	Flags joint applications
emi, emi_to_income	loan x 1000 / term; then / total income	Monthly repayment burden, a standard affordability ratio (assumes monthly income)
household_size, income_per_person	1 + married + dependents; income / size	Income available per household member
log_applicant_income, log_coapplicant_income, log_total_income, log_loan_amount, log_income_per_person	log(1 + value)	Compress the long right tail so extreme values stop dominating

3.4 Encoding

Column(s)	Technique	Why this technique
loan_status (target)	Label encoding (N = 0, Y = 1)	A target needs only two numbers
gender, married, education, self_employed	Binary 0/1 label encoding	Two categories: one column holds all the information; one-hot would add a duplicate
property_area (Rural, Semiurban, Urban)	One-hot encoding (Rural is the reference)	No natural order: codes 0/1/2 would wrongly suggest Urban is "twice" Semiurban
credit_history (Meets, Fails, Unknown)	One-hot encoding (Meets is the reference)	No order; keeps Unknown as its own signal
dependents	Kept as an ordinal number (0 to 3)	The order is real: 3 dependents is more than 1

One category per variable is dropped as the reference, because the remaining columns already imply it. Keeping all of them would make the columns perfectly redundant (the dummy variable trap), which harms linear models.

3.5 Outlier detection and treatment

Boxplots, the IQR rule (1.5 x IQR beyond the quartiles) and the z-score (more than 3 standard deviations) were applied to the training rows. The IQR rule flagged 4% to 8% of applications per variable (for example 40 applicant incomes and 33 loan amounts); the z-score flagged only 3 to 11, because extreme values inflate the standard deviation itself. These extremes are real applicants and the dataset is small, so **no rows were deleted**. Instead, a log transform removed most of the skew (applicant income 6.55 to 0.77; loan amount 2.79 to -0.18), and any remaining extreme value was capped at the IQR limits learned from the training rows (for example 27 applicant incomes, 39 loan amounts and 51 repayment ratios across the full data).

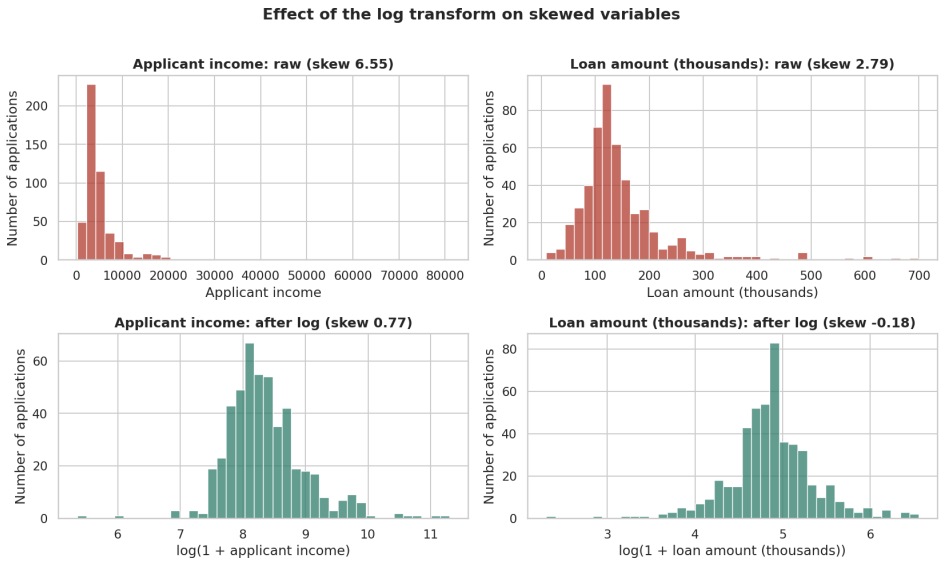


Figure 3. The log transform brings the skewness close to zero.

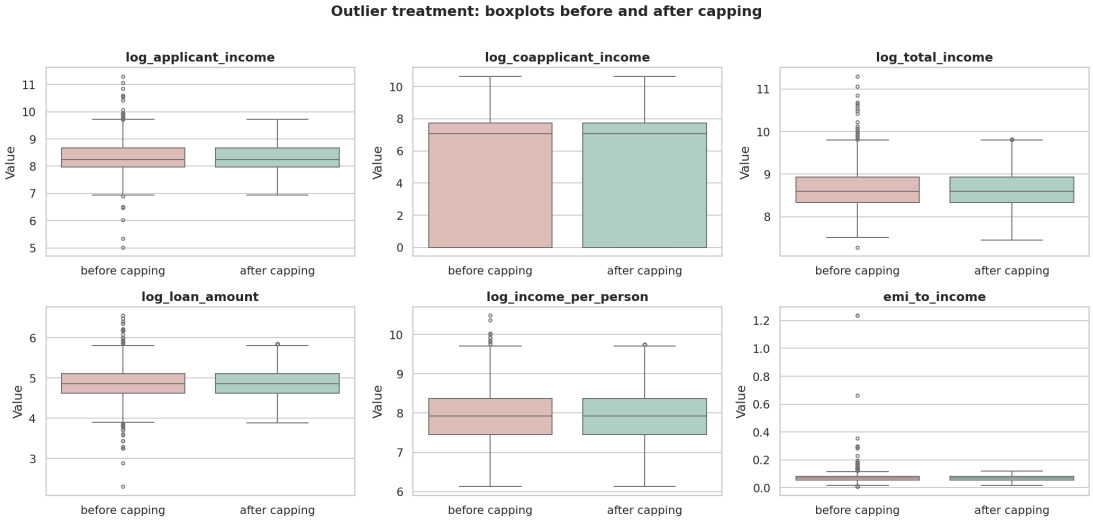


Figure 4. Boxplots before and after capping: the extreme points are limited, and no rows are removed.

3.6 Feature scaling

StandardScaler (mean 0, spread 1) was chosen for the continuous features. After the log transform and capping they are roughly symmetric with no extreme values, so standardising works well and suits likely next models (logistic regression, SVM, k-nearest neighbours). **MinMaxScaler** was rejected because a single extreme value sets the range and squeezes most applicants into a narrow band (mean 0.07 on raw income). **RobustScaler** is better when outliers remain, but they had already been treated. Scaling does not change the shape of a distribution, which is why the log transform and capping come first. The 0/1 columns are not scaled. The scaler was fitted on the training rows only.

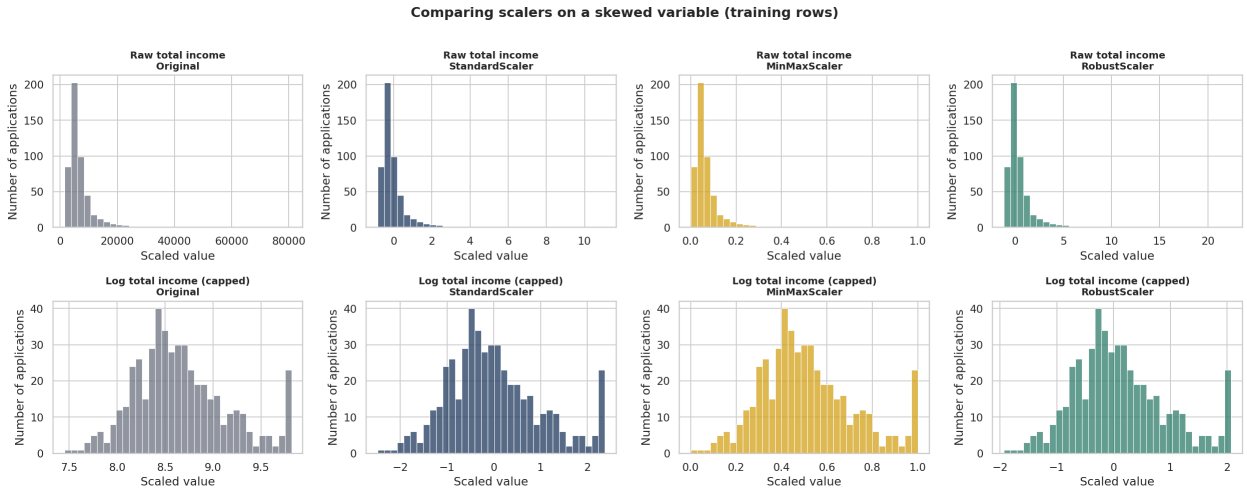


Figure 5. Three scalers compared on raw total income (top) and on the log-transformed, capped version (bottom).

4. Feature selection

Starting from 24 candidate features, three kinds of evidence were gathered on the training rows: correlation (to find redundancy), the relationship with the target (correlation, mutual information and a statistical test), and random forest importance. Because random forest importance favours continuous columns, ten random **noise probe** columns were used as a yardstick: a pure-noise continuous column scored about 0.06, and **every income and loan-amount feature scored no higher (0.05 to 0.07)**. Only credit history (importance 0.24, about 26 times the noise level) and property area clearly beat the noise.

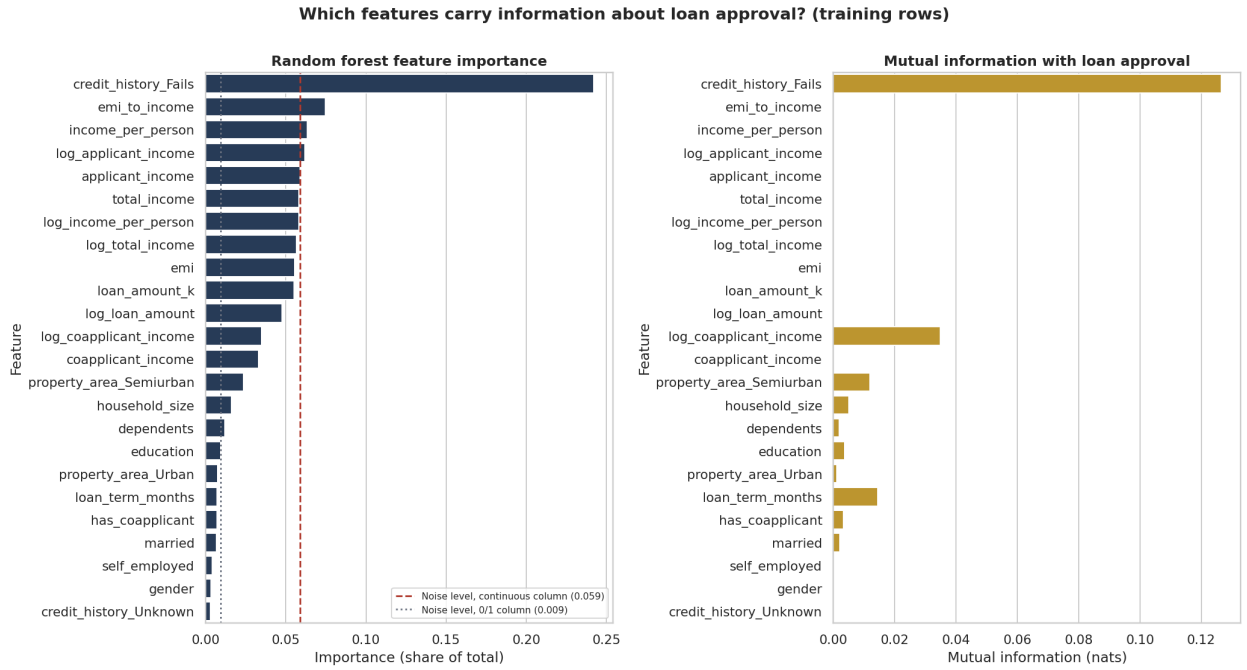


Figure 6. Random forest importance (with the noise-probe levels marked) and mutual information with loan approval.

Rules applied, in order

- **Superseded:** six skewed raw money columns were replaced by their capped log versions.
- **Redundancy (|r| >= 0.80):** the weaker feature of each pair was removed. Five pairs were found, including co-applicant income and its 0/1 flag (0.99) and household size and dependents (0.93).
- **Fairness:** gender and marital status were excluded because using them in credit decisions can discriminate. Neither is statistically supported (p = 0.84 and 0.18).
- **Evidence:** features with a statistically supported link to approval (p < 0.10) were kept: credit history, property area and education.

- **Domain-essential:** four features were kept despite weak sample evidence, chosen from lending logic before the results were seen: the three affordability measures (total income, loan amount, repayment ratio) and the "Unknown credit history" flag. 'Unknown' is not significantly different from 'Meets' (p = 0.175, only 43 applicants), but dropping it would silently treat applicants with no record as having a good one.
- **Group integrity:** all categories of a kept variable stay together, so no category is silently merged into the reference.

Selected feature	Reason kept
education	Supported (p = 0.074)
property_area_Semiurban	Supported (p = 0.001)
property_area_Urban	Kept with the other property-area category
credit_history_Fails	Strongest signal (correlation -0.51, p < 0.001)
credit_history_Unknown	Domain-essential: applicants with no credit record
emi_to_income	Domain-essential: repayment burden
log_total_income	Domain-essential: repayment capacity
log_loan_amount	Domain-essential: size of the request

Removed (16): raw skewed money columns (applicant_income, coapplicant_income, loan_amount_k, total_income, income_per_person, emi); redundant features (has_coapplicant, dependents); protected attributes (gender, married); and features with no supported link to approval (self_employed, loan_term_months, household_size, log_applicant_income, log_coapplicant_income, log_income_per_person). Income information is still carried by log_total_income. The largest correlation left among the selected features is 0.70, below the 0.80 limit.

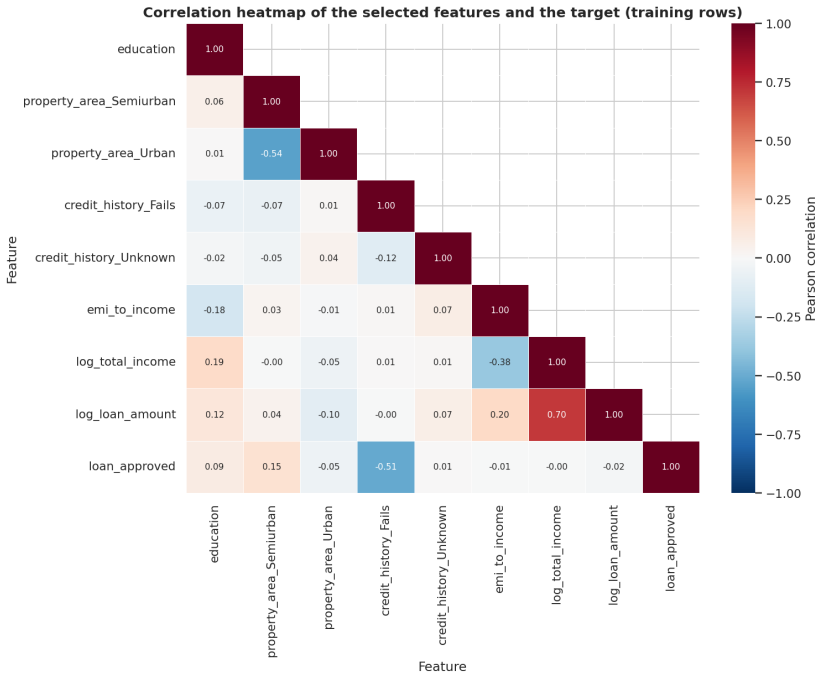


Figure 7. Correlation heatmap of the selected features and the target.

5. Validation: does the preprocessing help?

A plain logistic regression with 5-fold cross-validation on the training rows compared four versions of the data. This is a readiness check, not the modelling task.

Version	Accuracy	ROC-AUC	F1
A. Raw numeric columns only	0.796	0.643	0.869
B. All 24 engineered features	0.786	0.728	0.860
C. Selected (8 features)	0.798	0.732	0.870
D. Evidence-only (3 features)	0.798	0.742	0.870

Preparing the data lifts the ROC-AUC (how well applicants are ranked) by about 0.09 over the raw columns. Accuracy stays near 0.80 because credit history already carries it, and a rule that approves everyone scores 68.6%. Versions C and D differ by less than the cross-validation spread (0.02 to 0.04), so the four domain-essential features cost nothing measurable. Version B, which keeps every feature, does slightly worse, supporting the removal of redundant and noisy columns.

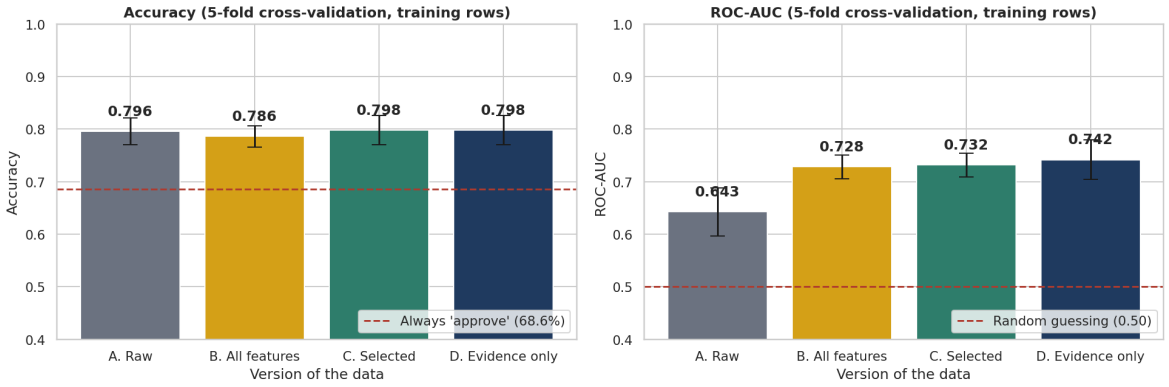


Figure 8. Cross-validated accuracy and ROC-AUC for the four versions of the data.

As a final readiness test, a model trained on the 491 training rows was scored once on the 123 hold-out rows: accuracy 85.4% (against 69.1% for always approving), ROC-AUC 0.854. It approved 84 of the 85 approvable applicants but rejected only 21 of 38 rejectable ones, so 17 applications that should have been rejected were approved. With only 123 rows this result is noisy and looks better than the cross-validated figures, so the cross-validation estimate is the steadier guide.

6. Final machine-learning dataset

File	Contents
data/processed/loan_cleaned.csv	614 rows. Readable cleaned data: missing values filled, new features added, before capping, encoding and scaling. Includes loan_id and the split flag.
data/processed/loan_ml_ready.csv	614 rows x 10 columns: 8 features (encoded, capped, scaled), the target loan_approved and a split flag (train / holdout).
loan_ml_ready_train.csv / _holdout.csv	The same data separated into the 491 training and 123 hold-out rows.
loan_kaggle_test_ml_ready.csv	The 367 unlabelled Kaggle test applications, processed with the same fitted pipeline (proof it works on new data).
preprocessing_parameters.json	Every fitted value: fill values, capping limits, scaler means, encodings and the final feature list.
feature_selection_summary.csv, ml_ready_data_dictionary.csv	The evidence and decision for each of the 24 candidates; a description of every final column.

Eight automatic checks all passed: no missing values, all columns numeric, no infinite values, no duplicate rows, target encoded as 0/1, no pair of features with |r| of 0.80 or more, the same row count as the cleaned dataset, and the scaler fitted on training rows only.

7. Key observations

- Credit history is the only strong predictor; everything else adds little on its own in this sample.
- Income and loan amount behave like random noise against the approval decision individually, yet remain necessary affordability measures for lending logic.

- Skewed monetary variables needed a log transform and capping; deleting outliers would have discarded real applicants.
- Accuracy is a poor yardstick here (69% by always approving), so future models should be judged on ROC-AUC, F1 and how well they catch rejections.
- Excluding gender and marital status costs no accuracy and avoids a fairness risk.

8. Justification summary and limitations

Decision	Why
Split before fitting anything	Prevents data leakage: the hold-out set never influences the preprocessing
Median for loan amount, mode for categories	Robust to skew and valid for categories, which have no average
"Unknown" category for credit history	Avoids inventing the strongest predictor; 'no record' is a real situation
Log transform plus IQR capping, no row deletion	Reduces skew while keeping genuine applicants in a small dataset
Binary encoding for two categories, one-hot for unordered ones	Avoids false ordering and duplicate columns
StandardScaler	Suits symmetric, capped features and the likely next models
Noise-probe yardstick	Random forest importance is biased towards continuous columns; the probe shows what counts as real
Fairness exclusions	Gender and marital status can discriminate in credit decisions

Limitations. The sample is small (614 rows), so estimates carry wide error bars. The target is a past approval decision, not a repayment outcome. Incomes are assumed to be monthly and the currency is not stated. Capping limits and the scaler come from 491 rows. Next step: train and compare baseline models on loan_ml_ready.csv using the same stratified split.